



Figure 1: Deployment design

The challenge of using immutable infrastructure

One of the core practices of DevOps is 'continuous delivery', wherein every single change is promoted from development to the production environment automatically. This sounds pretty fascinating and rather suitable for releasing a new version of an application, but in the real world, bringing it into practice is quite challenging, primarily when the enterprises fall under one of the following criteria.

- **Heterogeneous environments:** Here, applications are deployed on multiple environments (i.e., dev, stage, preprod or prod) or on more than one cloud (i.e., AWS, GCP or Azure).
- **Infrastructure auditing:** Lots of security tickets are received from SecOps routinely, which require immediate attention to fix the requested vulnerabilities.
- **Monolith applications:** The applications deployed on the cloud still follow the monolith design pattern, and decoupling them is nearly impossible at the moment.
- **Complex tooling:** The toolchain designed to promote changes from development to production environments is quite complex and requires manual intervention while making changes.

Since there are no clearly defined design practices to automate the deployment of immutable infrastructure, enterprises still prefer to manage the infrastructure in traditional ways, with the help of configuration management tools like Chef, Ansible, Puppet or SaltStack.

Tools	Description
Packer	Packer is a great tool to create identical machine images from a single source configuration for multiple platforms. For example, the pipeline depicted in Figure 1 uses Packer for baking the golden images from a base image.
Terraform	Terraform is an Infrastructure as Code tool that enables infrastructure provisioning on cloud and on-premise environments. Likewise, we use it to deploy cloud servers using the golden images.
Server-spec	Serverspec is an integration test framework that checks the servers' actual state and ensures they are provisioned correctly. It is used in the pipeline in Figure 1 for validating the infrastructure state.
Jenkins	Jenkins is an open source automation server that enables lots of integrations with the help of some great plugins. Jenkins is used in designing the end-to-end deployment pipeline described in Figure 1. Additionally, it allows a flawless integration of Packer, Serverspec and Terraform.

Solution: Automate immutable infrastructure deployments

Now that we have covered the core concepts of immutable infrastructure and the barriers to bringing it into practice, let us try and understand the patterns, processes, and practices needed to deploy it in any cloud environment.

Figure 1 illustrates the logical representation of the entire workflow, where the different phases of a pipeline are mentioned. The pipeline starts its execution by fetching the vanilla (base) machine image from the respective cloud repository. Then it pulls the latest application code commits, builds, and compiles. Upon successful validation, it creates a golden image from a running server, and publishes it to cloud repositories. And, finally, the golden image is deployed to the respective cloud environments.

The table above lists some excellent open source tools that can be used to implement the design workflow illustrated in Figure 1.

Deploying a demo application on AWS cloud

Now, let's implement the deployment design illustrated in Figure 1 and use it for launching an application instance on AWS.

The first step is to set up a Jenkins server and configure AWS credentials on it. Then, install Packer and Terraform binaries on the Jenkins server to launch the application instance in the AWS environment. Next, create a Jenkins job using the declarative pipeline given below:

```
pipeline {
  stages {
    // Pull latest code
    stage('Code Check-in') {
      steps {
        checkout([$class:
'GitSCM', branches: [[name: '*/
main']], userRemoteConfigs: [[url:
'https://github.com/vijayg92/immutable-
infrastructure']]])
        sh('ls -ltr')
      }
    }
  }
}
```